Multimedia Tools and Applications (2024) 83:79377–79396
https://doi.org/10.1007/s11042-024-18746-3

An audio encryption algorithm based on a non-degenerate 2D integer domain hyper chaotic map over GF(2^n)

Yafei Cao[1] · **Hongjun Liu**[1]

Received: 17 August 2023 / Revised: 14 February 2024 / Accepted: 24 February 2024 /
Published online: 5 March 2024
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract
To ensure the security transmission of audio signals in instant messaging tools, a chaos-based dual-channel one-time one-key audio encryption scheme was proposed. First, a non-degenerate 2D integer domain hyper chaotic map (2D-IDHCM) over GF(2^n) was constructed, the results of dynamics analysis showed that the 2D-IDHCM has ergodicity and large Lyapunov exponents. Then, based on 2D-IDHCM, two keyed strong S-Boxes were constructed. Further, an audio encryption & decryption algorithm was designed using 2D-IDHCM and keyed strong S-Boxes. The novelty includes: (1) 2D-IDHCM has no dynamic degradation, fast iteration speed and no need for data type conversion, which improves the efficiency of the algorithm. (2) The algorithm includes confusion process and diffusion process, which are resistant to most common attacks. Experimental results and security analysis demonstrated that the algorithm is sensitive to the initial condition, reduces the correlation of the original audio, and the total key space can reach 2^{224}. In addition, NPCR and UACI are all close to their ideal values of 99.6094070% and 33.4635070%, and the information entropy is close to the ideal value of 8.

Keywords Audio cipher · Integer domain hyper chaotic map · Galois field · S-Box · One-time one-key

1 Introduction

With the rapid development of real-time communication technology, speech signal can be transmitted over insecure network [1, 2], therefore, the issue of audio security has been widely concerned. In audio transmission, if the audio information is not processed, there may be a risk of privacy disclosure [3, 4]. A good cryptosystem should have a large key space and be sensitive to the initial condition. To ensure the security transmission of audio signals in instant messaging tools, some audio encryption methods have been proposed. Shah et al. [5] proposed a 3D chaotic map and applied it to digital audio encryption. Fu et al. [6] constructed a 4D hyper chaotic system based on the foundation

✉ Hongjun Liu
sms_liuhj@ujn.edu.cn

[1] School of Mathematical Sciences, University of Jinan, Jinan 250022, Shandong, China

of a 3D Lü chaotic system and applied it to audio encryption. The results show that the audio encryption scheme based on hyper chaotic system is feasible.

In recent years, chaotic maps have been widely used in audio encryption algorithms due to its pseudo randomness, unpredictability, ergodicity and sensitivity to initial values [7, 8]. For a chaotic map, its sensitivity to initial conditions enables it to produce good pseudo-random time sequences, which can then be applied to cryptography as key stream. Liu et al. [9] proposed an audio encryption scheme by confusion and diffusion based on multi-scroll chaotic system and one-time keys, and enhanced the security of the encryption algorithm through a nonlinear equation. Wu et al. [10] designed an audio encryption algorithm based on a nested chaotic system. The encryption algorithm generates the key stream through 2D-LNIC, which is a process of scrambling and diffusion simultaneously. Rahul et al. [11] introduced a chaos-based audio encryption algorithm using biometric image and SHA-256 hash algorithm. Liu et al. [12] proposed an audio block encryption using 3D chaotic system with adaptive parameter perturbation. The proposed scheme has the characteristic of flexible block size, fast encryption & decryption algorithm, and sensitivity to tiny change of plain audio.

In the process of algorithm design, the nonlinear characteristics have an important impact on the security of cryptographic algorithm. The S-Box serves as the only nonlinear component in many cryptographic algorithms. The S-Box with high nonlinearity can improve the security strength of cryptographic algorithm. S-Box is usually designed using algebraic structures and methods based on chaotic map, and is widely used in symmetric ciphers, such as RC4, AES, ZUC and SM4 [13]. However, there have internal weaknesses, such as fixed point, reverse fixed points, or short period rings, which can be attacked by the cryptanalyst. Hence, it is necessary to construct a strong S-Box without weaknesses.

In recent years, researchers have constructed many strong S-Boxes with high nonlinearity through chaotic maps. Si et al. [14] constructed a keyed strong S-Box with higher nonlinearity based on 2D hyper chaotic map and algebraic operations. Based on 2D chaotic map, Liu et al. [15] derived a recursive formula for calculating the number of S-Boxes and designed a strong S-Box construction algorithm.

Due to the finite precision effect of the digital platform [16], the iterative process of any chaotic map will be affected by truncation error and rounding error, which could inevitably lead to dynamic degradation, such as short cycle length, correlation and low linear complexity [17, 18]. To solve the problem, the discrete time chaotic maps on integer domain have been constructed. Wang et al. [19] proposed a method for constructing chaotic systems on integer domain, and proved the existence of Devaney's definition of chaos. Bashier et al. [20] introduced two chaotic maps over integer domain and designed an image encryption algorithm. Therefore, it is important to study the application of integer field chaotic map in cryptography.

To ensure the secure transmission of audio data over insecure network, we will propose a chaos-based dual-channel one-time one-key audio encryption scheme. First, we will construct a 2D-IDHCM over GF(2^n) with ergodicity and large Lyapunov exponents, and analyze its dynamical behaviors. Then, we will design two keyed strong S-Boxes based on 2D-IDHCM, and eliminate the weaknesses of S-Boxes through nine-grid local swapping and merging short period rings method [21]. Further, we will use the sequences generated by 2D-IDHCM and keyed strong S-Boxes to XOR and rearrange the audio data respectively. The main contributions and novelty of this paper are as follows:

(1) To resist the dynamic degradation caused by chaotic map in digital domain, we constructed a 2D-IDHCM over GF(2^n) and analyzed its dynamical behavior.
(2) The initial condition of 2D-IDHCM relies on the hash value of true random environmental noise, which can help generate different initial conditions, making the chaotic trajectory more unpredictable.
(3) Based on 2D-IDHCM, we designed two keyed strong S-Boxes without fixed point, reverse fixed points and short period rings.
(4) We designed a one-time one-key audio encryption algorithm using 2D-IDHCM and S-Boxes, and analyzed the experimental results to evaluate its security and effectiveness.

The novelty is that (1) 2D-IDHCM has no dynamic degradation, fast iteration speed and no need for data type conversion, which improves the efficiency of the algorithm. (2) The algorithm includes two processes: confusion and diffusion. The former is to arrange the positions of elements in the two channels with two keyed strong S-Boxes, and the latter is to use the integer sequences generated by 2D-IDHCM to change their values.

The remaining sections are structured as follows: Section 2 is to construct a 2D-IDHCM and analyze the dynamic characteristics. Section 3 designs an audio encryption and decryption algorithm. Section 4 is experimental results. Section 5 conducts security and performance analysis. Section 6 reaches a conclusion.

2 Construct a 2D-IDHCM

A 2D-IDHCM can be defined as Eq. (1).

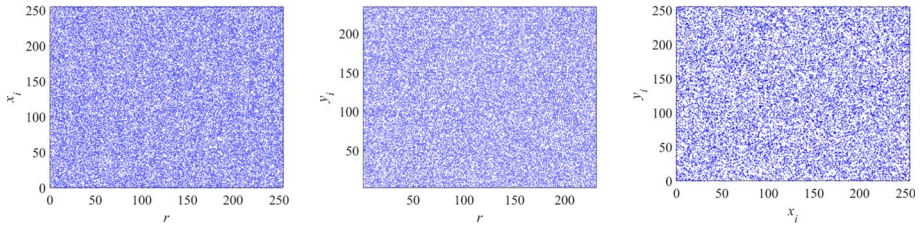
$$\begin{cases} x_{i+1} = ra_{11} \times \mathrm{gf}\left(x_i, n, m\right) + a_{12} \times \mathrm{gf}\left(y_i, n, m\right) + 5 \\ y_{i+1} = a_{21} \times \mathrm{gf}\left(x_i, n, m\right) + ra_{22} \times \mathrm{gf}\left(y_i, n, m\right) \end{cases}, \tag{1}$$

where function gf(*x, n, m*) represents a Galois field array from the matrix x, the Galois field has 2^n elements, $n \in \{8,16\}$, and m is the integer representation of the primitive polynomial. The state variables x_i, y_i are integers within the interval $[0, 2^n - 1]$, control parameter and four coefficients r,a_{11},a_{12},a_{21},a_{22} are all integers within $[2, 2^n - 1]$.

Theoretically, the exponent n can be an integer within the interval $[2, +\infty)$, however, arising from the limited precision of platform, here we set $n \in [2, 16]$. Generally, we set $n = 8$ or $n = 16$ in 2D-IDHCM, the generated integers can be expressed as one byte or two bytes, which can be easily processed by a computer.

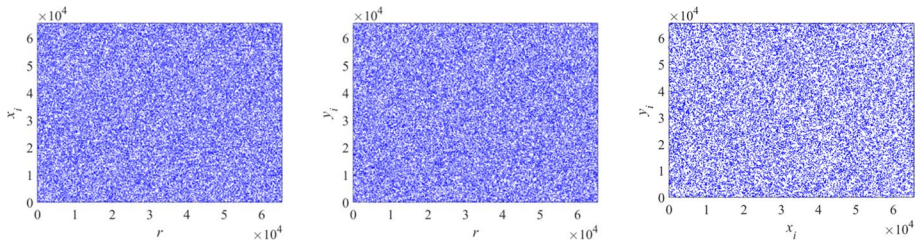
Compared with digital domain chaotic map [17, 18], the constructed integer domain chaotic map has the following advantages: (1) the iteration speed is fast and no data type conversion is required, (2) the 2D-IDHCM overcomes the limited precision effect of the digital platform and has no dynamic degradation, and (3) the sequences generated by 2D-IDHCM iteration have high nonlinearity.

In Section 3.2 of Ref. [22], we proved that 2D-IDHCM satisfies Devaney's definition of chaos, including: (1) the sequences generated by 2D-IDHCM are sensitive to the initial values, and any small initial error can produce large changes after several iterations, (2) topological transitivity means that the entire metric space will be traversed under the action of mapping, and (3) the density of periodic points indicates that 2D-IDHCM has strong determinism and regularity. In addition, in Section 3.3, we proved that Lyapunov exponent is bounded.



(a) Bifurcation diagram (r-x_i) (b) Bifurcation diagram (r-y_i) (c) Phase diagram (x_i-y_i)

Fig. 1 Bifurcation diagrams and phase diagram ($n = 8$)



(a) Bifurcation diagram (r-x_i) (b) Bifurcation diagram (r-y_i) (c) Phase diagram (x_i-y_i)

Fig. 2 Bifurcation diagrams and phase diagram ($n = 16$)

In the following, we analyzed the dynamic characteristics of 2D-IDHCM, such as bifurcation diagram, phase diagram, Lyapunov exponent, correlation dimension, sample entropy.

From the bifurcation diagrams shown in Figs. 1 and 2, we can infer that the polynomial state variables x_i and y_i change with the control parameter r randomly. And from the phase diagrams shown in Figs. 1 and 2, we can infer that the trajectory distribution of state points generated by 2D-IDHCM have ergodicity in phase space.

2.1 Lyapunov exponent (LE)

LE is a measure of the average exponential rate of convergence or divergence between adjacent trajectory in phase space, which can reflect the complexity of chaotic map to a certain extent. From the LEs shown in Fig. 3, we can infer that 2D-IDHCM has two positive LEs, which keep increasing with the varying control parameter r. Hence, the 2D-IDHCM is non-degenerate and exhibits hyper chaotic behavior within a large parameter range.

2.2 Correlation dimension (CD)

CD can reflect the number of variables required for a set of discrete state points to move in space, and quantitatively describe the geometric characteristics of chaotic attractors [23]. Generally, the CD can be used to directly judge whether a time sequence has chaotic characteristics, and it is also an important statistic to detect whether the data has nonlinearity using alternative data method.

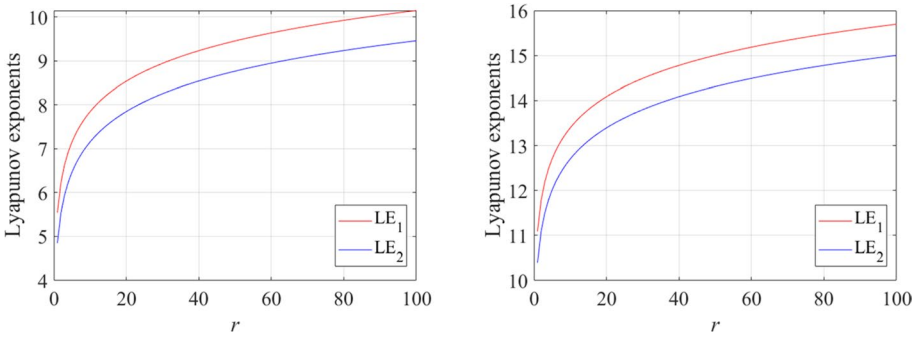


Fig. 3 Lyapunov exponents

From the CDs shown in Fig. 4, we can infer that the state points occupy a higher dimension in the phase space. In addition, some CD values are greater than 2, which indicates the trajectory is very complex and irregular, and may not be fully described with a 2D coordination system even if it is a 2D map [24].

2.3 Sample entropy (SE)

SE can measure the complexity of time series. The lower the value of SE, the higher the self-similarity of the sequences. The larger the SE, the more complex the time series [25]. The SE has two advantages: (1) the calculation of the SE does not depend on data length, and (2) the SE has better consistency, i.e., the change of vector length and similarity measure have the same influence on the SE.

From the SEs shown in Fig. 5, we can infer that under a certain control parameter r, the oscillation amplitude of the polynomial state variables x_i and y_i are larger than 2, which indicates that 2D-IDHCM over gf(x,n,m) has more complex trajectories and can be used to generate time series with good randomness.

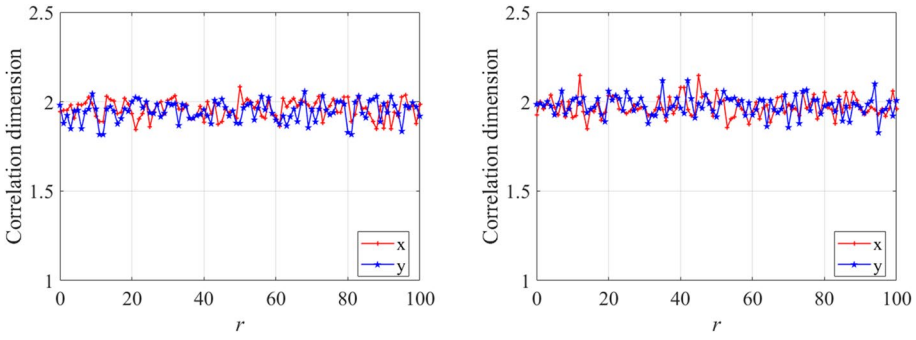
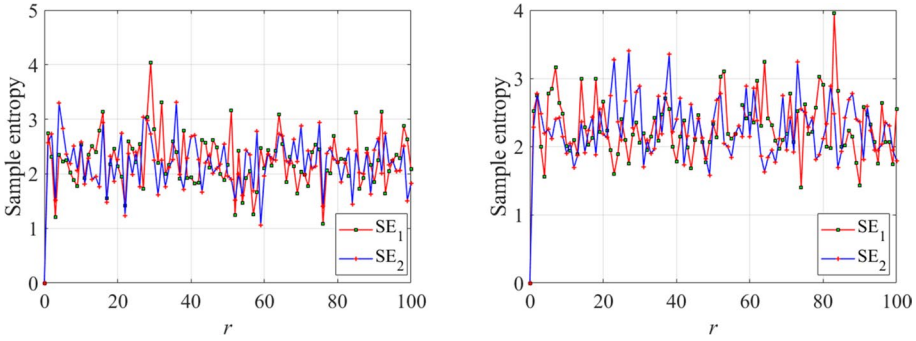


Fig. 4 Correlation dimension



(a) n=8 (b) n=16

Fig. 5 Sample entropy

3 Audio encryption and decryption algorithm

Due to the finite precision effect of computer, an audio encryption algorithm based on chaotic map in digital domain will inevitably lead to dynamic degradation [12, 26]. Therefore, it is necessary to design an audio encryption algorithm based on integer domain hyper chaotic map over GF(2^n).

The encryption algorithm mainly consists of three steps:

(1) Construct two keyed strong S-Boxes. The integer sequences $\{x_i\}_s$ and $\{y_i\}_s$ generated by 2D-IDHCM were used to construct two keyed strong S-Boxes without fixed point, reverse fixed points and short period rings. In addition, the initial condition of 2D-IDHCM is derived from true random environmental noise, which helps to produce completely different initial conditions.

(2) Changes the size of the element values in the left and right channels. First, the values in the audio file are enlarged 10^5 times, so that they are divided into integer part and decimal part. Then, we use integer sequences $\{x(i)\}$ and $\{y(i)\}$ to XOR integer part. Further, the processed numbers are reduced by 10^5 times to get the double precision floating point number within the interval [-1,1].

(3) Rearrange the positions of elements in the two channels with two keyed strong S-Boxes.

The proposed method can be illustrated by a block diagram, as shown in Fig. 6.

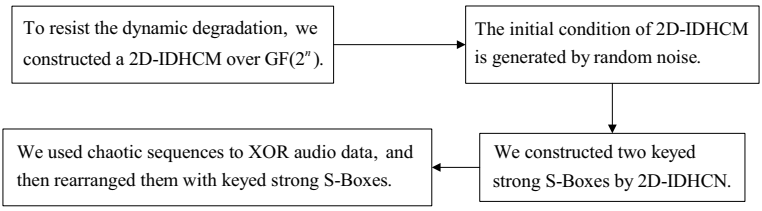


Fig. 6 A block diagram of the proposed method

3.1 Constructing two keyed strong S-Boxes

Input: initial condition $(x_1,y_1,r)_s$.

Output: two keyed strong S-Boxes S_L and S_R.

In this part, we transform the true random environmental noise N into the initial condition of 2D-IDHCM, iterate the Eq. (1), generate the integer sequences $\{x_i\}_s$ and $\{y_i\}_s$, and then construct two keyed strong S-Boxes. The detailed steps are as follows.

Step 1. Calculate the hash value H of the true random noise signal N using Eq. (2).

$$H = \text{hash}_{256}(N), \tag{2}$$

where function $\text{hash}_{256}()$ represents a public hash algorithm, such as SHA2-256, or SHA3-256. SHA2-256 and SHA3-256 satisfy antigenic image attack, resist second preimage attack, and resist strong collision attack, and are all one-way hash functions. The noise signal N is randomly sampled from true random environmental noise [8], which helps to produce completely different initial conditions, ensuring that the encryption scheme is one-time one-key. H can be expressed as hexadecimal format $H=\{h_1, h_2, h_3,\ldots, h_{64}\}$. In summary, this step is mainly to transform the true random environmental noise N into the hexadecimal hash value H.

Note: Random noise N is a kind of noise caused by the accumulation of a large number of fluctuations generated randomly in time, whose value cannot be predicted in a given instant. Here the noise signal N is sampled from true random environmental noise, such as running water, rain, wind and so on. The sampled random noises are different under different time and place, which can ensure a one-time one-key encryption scheme.

Step 2. Using Eq. (3) to convert H into initial condition $(x_1, y_1, r)_s$ of 2D-IDHCM to construct the S-Box.

$$\begin{cases} x_1 = 5 + (\text{hex2dec}(H(1:24)))\text{mod}(2^n - 6) \\ y_1 = 7 + (\text{hex2dec}(H(21:44)))\text{mod}(2^n - 8) \\ r = 11 + (\text{hex2dec}(H(41:64)))\text{mod}(2^n - 12) \end{cases}, \tag{3}$$

where function hex2dec() converts a hexadecimal number to a decimal number, and function mod(a,m) returns the remainder after division of a by m, where a is the dividend and m is the divisor. This step is to segment the 256-bit hash value H and convert it into decimal numbers as the initial condition $(x_1, y_1, r)_s$ of 2D-IDHCM, and then iterate the Eq. (1) to generate chaotic sequences.

Step 3. Iterate the Eq. (1) 200 times with $(x_1, y_1, r)_s$ to eliminate the influence of the transient process, then continue to iterate it M times to generate the integer sequences $\{x_i\}_s\Big|_{i=1}^{M}$ and $\{y_i\}_s\Big|_{i=1}^{M}$. Generally, $M \geq 600$. Then, transform the sequence $\{x_i\}_s$ to integer sequence$\{S_L(i)\} \in [0,255]$, and remove the duplicate elements in $\{S_L(i)\}$, as shown in Eq. (4). If the number of elements of $\{S_L(i)\}$ is less than 256, then, set $M=M+100$ and repeat Step 3 until $S'_L = \{S_L(i)\}$ with 256 non-repeatable elements is obtained.

In addition, we use the same method to transform the sequence $\{y_i\}_s$ to integer sequence $\{S_R(i)\} \in [0,255]$ using Eq. (4), repeating Step 3 until $S'_R = \{S_R(i)\}$ with 256 non-repeatable elements is generated. This step focuses on constructing two candidate S-Boxes S'_L and S'_R with chaotic sequences $\{x_i\}_s$ and $\{y_i\}_s$.

$$\begin{cases} \{S_L(i)\} = \text{unique}\left(\{x_i\}_s \bmod 256\right) \\ \{S_R(i)\} = \text{unique}\left(\{y_i\}_s \bmod 256\right) \end{cases}. \tag{4}$$

Step 4. Eliminate the possible fixed point, reverse fixed points and short period rings [21] in S'_L. Further, obtain a keyed strong S-Box S_L. Similarly, a keyed strong S-Box S_R can be constructed using $\{y_i\}_s$ in the same way. This step is to transform two candidate S-Boxes S'_L and S'_R into two keyed strong S-Boxes S_L and S_R.

Through Steps 1–4, two keyed strong S-Boxes S_L and S_R are generated with sequences $\{x_i\}_s$ and $\{y_i\}_s$ respectively, which are applied in Step 7 of of Section 3.3 for rearranging the positions of elements in the audio file.

3.2 Generating initial condition of 2D-IDHCM

In this section, we rearrange the 256-bit hash value H to get H_1, then convert H_1 to the initial condition of 2D-IDHCM, and generate the integer sequences $\{x(i)\}$ and $\{y(i)\}$. The detailed steps are as follows.

Step 1. Re-arrange H, divide H into an 8×8 array [11], then traverse each element in a Zigzag fashion to get $H_1 = \{h_1, h_2, h_9, \ldots, h_{64}\}$, as shown in Fig. 7. This process is to change the initial condition of the 2D-IDHCM to generate different chaotic sequences.

Step 2. Convert H_1 into the initial condition $(x_1, y_1, r)_{\oplus}$ of 2D-IDHCM using Eq. (5), and the sequences generated by the 2D-IDHCM are used for XOR audio data. In this step, the hash value H_1 is segmented, converted to decimal numbers, and then added as the initial condition for 2D-IDHCM.

$$\begin{cases} x_1 = 3 + \left(\text{hex2dec}\left(H_1(1:12)\right) + \text{hex2dec}\left(H_1(10:21)\right)\right) \bmod (2^n - 4) \\ y_1 = 7 + \left(\text{hex2dec}\left(H_1(20:31)\right) + \text{hex2dec}\left(H_1(29:40)\right)\right) \bmod (2^n - 8). \\ r = 13 + \left(\text{hex2dec}\left(H_1(41:52)\right) + \text{hex2dec}\left(H_1(53:64)\right)\right) \bmod (2^n - 14) \end{cases} \tag{5}$$

3.3 Encryption algorithm

The encryption algorithm includes confusion process and diffusion process. The former is to arrange the positions of the elements in the two channels with two keyed strong S-Boxes,

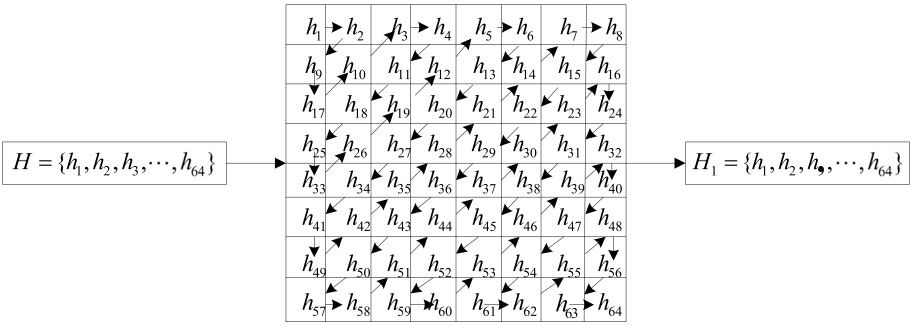


Fig. 7 Zigzag transformation

and the latter is to use the integer sequences $\{x(i)\}$ and $\{y(i)\}$ generated by 2D-IDHCM to change their values. The purpose is to transform the plain audio with distinct time domain waveform into a cipher audio with relatively uniform and random distribution to resist against common attacks [12].

Input: The plain dual-channel audio file A_p with the size of $L_A \times 2$, initial conditions $(x_1,y_1,r)_s$ and $(x_1,y_1,r)_{\oplus}$.

Output: The ciphered audio A_C with the same size.

First, through Steps 1–6, we changed the size of the element values in the two channels. Then, Step 7 is to rearrange the positions of the elements through the order of two keyed strong S-Boxes. The detailed measures are as follows.

Step 1. Read the dual-channel audio file A_P to obtain the L_A-by-2 matrix A, where L_A is the number of audio samples read and 2 is the number of audio channels in the file [9], and sample frequency FS in Hertz.

Step 2. The numbers in matrix A are double-precision floating-point number within the interval [-1,1], then the numbers in matrix A are expanded by 10^5 times, obtain the integer part A_I and decimal part A_F using Eq. (6). This process mainly enlarges the numbers in matrix A to get the integer part and the decimal part.

$$\begin{cases} A_I = \mathrm{fix}\left(\mathrm{abs}(A) \times 10^5\right) \\ A_F = \mathrm{rem}\left(A \times 10^5, 1\right) \end{cases}, \tag{6}$$

where function abs(α) represents taking the absolute value of α, function fix(α) rounds each element of α to the nearest integer toward zero, and function rem(a, b) returns the remainder after division of a by b, where a is the dividend and b is the divisor.

Step 3. By the method described in Section 3.1, construct two keyed strong S-Boxes S_L and S_R based on 2D-IDHCM with $(x_1,y_1,r)_s$.

Step 4. Iterate Eq. (1) L_A times with $(x_1,y_1,r)_{\oplus}$, generating two integer sequences $\{x(i)\}\Big|_{i=1}^{L_A}$ and $\{y(i)\}\Big|_{i=1}^{L_A}$. In addition, the initial condition of 2D-IDHCM is described in Section 3.2. The main purpose of this step is to generate the integer sequences to XOR integer part A_I.

Step 5. Divide matrix A_I into left and right channels A_I^L and A_I^R, and XOR them with sequences $\{x(i)\}\Big|_{i=1}^{L_A}$ and $\{y(i)\}\Big|_{i=1}^{L_A}$ respectively using Eq. (7) to obtain an updated A_I. This step mainly involves changing the size of the integer part A_I values obtained in Step 2 of Section 3.3.

$$\begin{cases} A_I^L = A_I^L(i) \oplus x(i) \\ A_I^R = A_I^R(i) \oplus y(i) \end{cases}, i = 1, 2, \cdots, L_A, \tag{7}$$

where $\bigoplus$ is a bit-wise exclusive or (XOR) operator.

Step 6. Transform A_I and A_F into A_C within the interval [-1,1] using Eq. (8). This step is mainly to add the updated integer part and the decimal part, reduce by 10^5 times, and restore to the double precision floating point number within [-1,1].

$$A_C = \left(\left(\mathrm{sign}\left(A_F\right) \times A_I\right) + A_F\right) \times 10^{-5}, \tag{8}$$

where function sign(β) returns an array with same size as β, "1" "-1" "0" means that the corresponding element in β is greater than 0, less than 0, and equal to 0, respectively. In summary, the main purpose of the above steps is to change the size of the audio data values.

Step 7. Divide matrix A_C into $L_A/256$ segments with the same length of 256 using Eq. (9), re-arrange the left channel in the order of S_L-Box and the right channel in the order of S_R-Box using Eq. (10), to obtain the ciphered audio file A_C. This step mainly involves rearranging the positions of elements in audio file A_C with the order of two keyed strong S-Boxes S_L and S_R.

$$\begin{cases} A_C^L(1:256, 1:L_A/256) = \text{reshape}(A_C^L, 256, L_A/256) \\ A_C^R(1:256, 1:L_A/256) = \text{reshape}(A_C^R, 256, L_A/256) \end{cases}, \tag{9}$$

$$\begin{cases} A_C^L(i, 1:L_A/256) = A_C^L(S_L(i-1), 1:L_A/256) \\ A_C^R(i, 1:L_A/256) = A_C^R(S_R(i-1), 1:L_A/256) \end{cases}, i = 1, 2, \cdots, 256, \tag{10}$$

where function reshape() means to reconstruct the array.

The flowchart of the encryption algorithm is shown in Fig. 8.

3.4 Decryption algorithm

The decryption algorithm includes two processes: confusion and diffusion. (1) The confusion process was completed through Steps 1–4. We use two inverse S-Boxes to rearrange the positions of elements in the cipher audio file A_C. (2) The diffusion process was completed through Steps 5–7. We change the size of the values in both channel elements. The detailed steps are as follows.

Input: The cipher audio file A_C, initial conditions $(x_1,y_1,r)_s$ and $(x_1,y_1,r)_{\oplus}$.

Output: The decrypted audio A_P.

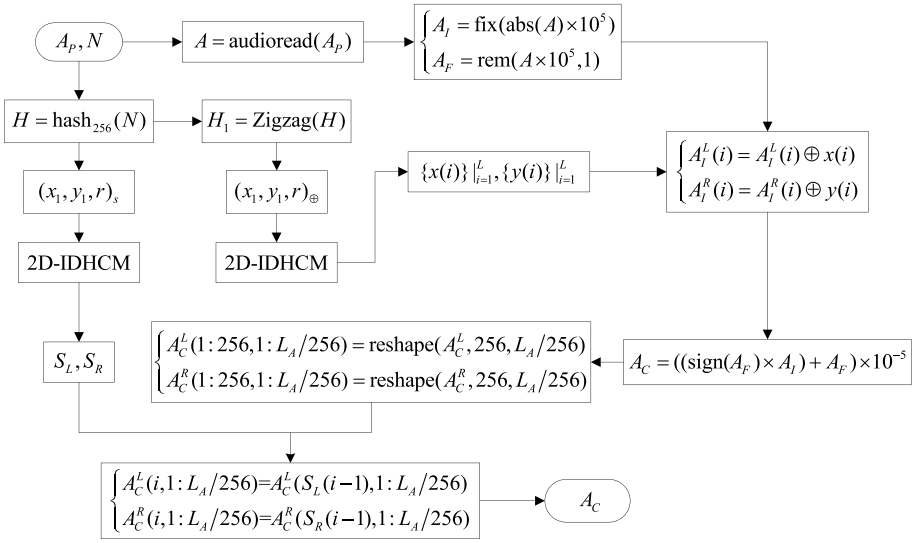


Fig. 8 Flowchart of encryption algorithm

Step 1. Read the dual-channel audio file A_C to obtain L_A-by-2 matrix A_C, where L_A is the number of audio samples read and 2 is the number of audio channels in the file, and sample frequency FS in Hertz.

Step 2. Construct two keyed strong S-Boxes S_L, S_R with $(x_1,y_1,r)_s$, then construct inverse S-Boxes S_{IL}, S_{IR}.

Step 3. Iterate Eq. (1) L_A times with $(x_1,y_1,r)_{\oplus}$, to generate two integer sequences $\{x(i)\}\big|_{i=1}^{L_A}$ and $\{y(i)\}\big|_{i=1}^{L_A}$ to XOR audio data.

Step 4. Divide matrix A_C into $L_A/256$ segments with the same length of 256 using Eq. (11). Re-arrange the left and right channels in the order of the inverse S-Boxes S_{IL}, S_{IR}, using Eq. (12). This step mainly restores the original order of numbers in the matrix A_C using inverse S-Boxes S_{IL} and S_{IR}.

$$\begin{cases} A_C^L(1:256,1:L_A/256) = \text{reshape}\left(A_C^L,256,L_A/256\right) \\ A_C^R(1:256,1:L_A/256) = \text{reshape}\left(A_C^R,256,L_A/256\right) \end{cases}, \tag{11}$$

$$\begin{cases} A_C^L(i,1:L_A/256) = A_C^L\left(S_{IL}(i-1),1:L_A/256\right) \\ A_C^R(i,1:L_A/256) = A_C^R\left(S_{IR}(i-1),1:L_A/256\right) \end{cases}, i = 1,2,\cdots,256. \tag{12}$$

Step 5. The numbers in matrix A_C are double-precision floating-point number within the interval [-1,1], then the numbers in matrix A_C are expanded by 10^5 times, so obtain the integer part A_I and decimal part A_F using Eq. (13). This process mainly enlarges the numbers in matrix A_C to get the integer part and decimal part.

$$\begin{cases} A_I = \text{fix}\left(\text{abs}\left(A_C\right)\times 10^5\right) \\ A_F = \text{rem}\left(A_C\times 10^5,1\right) \end{cases}. \tag{13}$$

Step 6. Divide matrix A_I into left and right channels A_I^L andA_I^R, and XOR them with sequences $\{x(i)\}\big|_{i=1}^{L_A}$ and $\{y(i)\}\big|_{i=1}^{L_A}$ respectively using Eq. (14), to obtain updated A_I. This step mainly involves changing the size of the integer part A_I values obtained in Step 5 of Section 3.4.

$$\begin{cases} A_I^L(i) = A_I^L(i)\oplus x(i) \\ A_I^R(i) = A_I^R(i)\oplus y(i) \end{cases}, i = 1,2,\cdots,L_A. \tag{14}$$

Step 7. Transform A_I and A_F into A_C within the interval [-1,1] using Eq. (15), then obtain the plain audio file A_P. This step is mainly to add the updated integer part and the decimal part, reduce by 10^5 times, and restore to the double precision floating point number within [-1,1]. In summary, Steps 5–7 are to change the size of the audio data values.

$$A = \left(\left(\text{sign}\left(A_F\right)\times A_I\right)+A_F\right)\times 10^{-5}. \tag{15}$$

The flowchart of the decryption algorithm is shown in Fig. 9.

4 Experimental results

The time domain characteristic charts of audio file Sample1.wav and Sample2.wav are shown in Fig. 10, from which we can infer that the amplitude distribution of the ciphered audio is uniform.

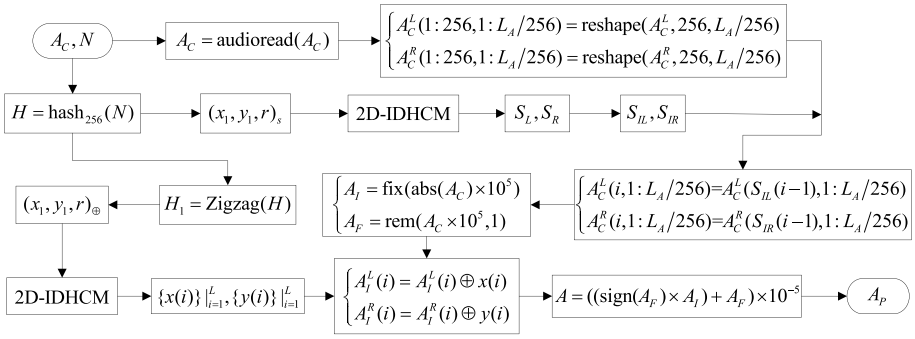
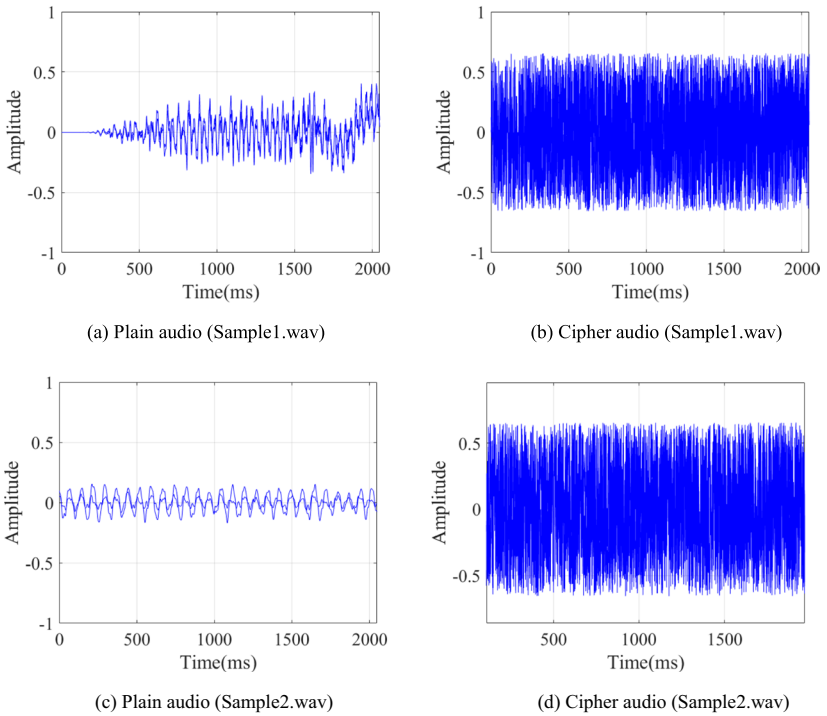


Fig. 9 Flowchart of decryption algorithm

The frequency distribution charts of plain and the ciphered audio are given in Fig. 11, from which we can infer that the frequency of the ciphered audio is uniformly distributed, which can improve the security of the encryption algorithm.

In addition, two keyed strong S-Boxes S_L and S_R without weaknesses are listed in Tables 1 and 2.



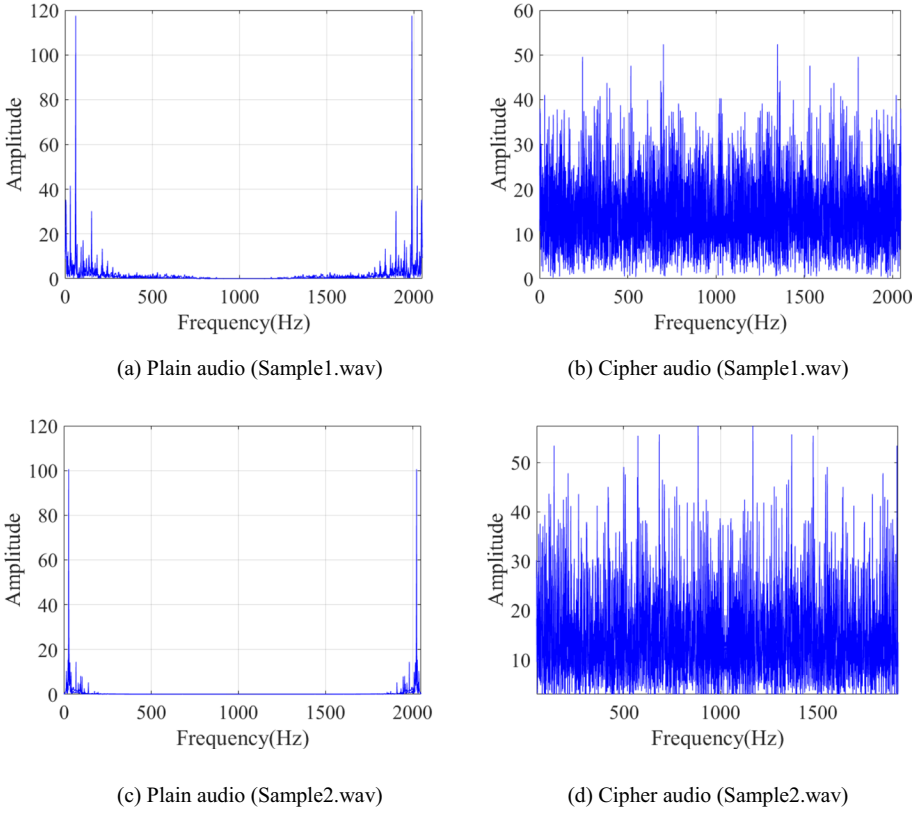
(a) Plain audio (Sample1.wav)

(b) Cipher audio (Sample1.wav)

(c) Plain audio (Sample2.wav)

(d) Cipher audio (Sample2.wav)

Fig. 10 Time domain characteristic charts



(a) Plain audio (Sample1.wav)

(b) Cipher audio (Sample1.wav)

(c) Plain audio (Sample2.wav)

(d) Cipher audio (Sample2.wav)

Fig. 11 Frequency distribution charts

Table 1 A strong S-Box S_L

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	30	AC	FA	1A	39	95	BC	0C	CF	E7	65	20	D6	03	DD	78
1	E4	3E	F2	C8	80	44	5D	48	B0	10	82	2D	AD	9D	C9	97
2	72	5F	86	E3	59	0B	0F	C1	35	2E	A3	4F	D7	64	0E	A6
3	E6	F0	A1	81	84	0A	40	62	A9	53	3D	FC	9E	C7	89	F1
4	C6	52	5A	2F	C4	9C	09	66	C0	F5	AF	F4	27	16	25	D8
5	76	91	F8	D4	6A	DC	B8	68	69	75	38	6F	D2	CD	29	63
6	7B	EA	EC	4B	19	90	15	2B	42	12	EB	2C	6D	1C	60	67
7	00	74	73	55	5B	17	A8	93	AB	6E	B5	1E	F7	61	A7	23
8	7F	83	4A	7E	43	CB	3B	EF	45	8E	96	B2	DE	D1	9F	A0
9	57	F3	08	01	13	3A	9B	4C	99	41	05	7A	8D	3F	32	49
A	C3	5C	CE	FE	31	E0	54	88	8C	58	E1	D5	0D	26	71	F6
B	06	21	B1	D0	2A	BB	CC	E9	E2	9A	92	7C	CA	6C	1F	37
C	D3	34	EE	22	18	50	94	98	FD	7D	4D	47	A2	8F	56	33
D	DB	DF	11	E5	07	B4	36	51	DA	79	ED	4E	8A	FF	14	D9
E	B6	24	28	04	BA	B9	A5	E8	BF	02	1D	F9	B3	3C	C5	FB
F	C2	BD	B7	A4	85	8B	BE	77	AE	1B	5E	46	70	87	AA	6B

Table 2 A strong S-Box S_R

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	EC	F2	E5	47	8F	48	9D	D7	49	C9	6D	02	30	3E	78	5D
1	C2	25	52	E3	D3	13	06	6A	F1	7F	AE	8D	B5	A7	0B	B6
2	3D	E2	B4	C0	D8	68	3F	2C	E7	89	DC	01	99	BE	EF	5B
3	65	D0	0C	45	72	41	28	BD	46	DB	D6	2E	1A	B3	00	FE
4	31	59	AB	83	EB	AA	7A	A0	F8	F9	AD	17	5C	A5	F6	8C
5	E1	1D	40	4F	6E	98	D9	3C	85	67	1E	DA	A6	6C	E0	93
6	91	56	8A	75	81	0E	B8	32	F0	2B	14	C6	4C	F4	71	C1
7	BF	BB	29	95	05	18	7D	53	B7	69	58	FF	39	8B	12	BA
8	86	FD	F7	B9	1C	F5	61	E8	9B	43	D5	87	9A	24	EA	55
9	10	9C	F3	82	94	51	4A	84	CD	04	A4	A2	36	E4	35	03
A	64	33	5F	1F	8E	27	B0	6F	CC	63	77	7C	34	5A	16	3B
B	76	2F	4E	09	4B	07	57	DE	CF	CA	C3	42	D1	DD	23	08
C	B2	2D	97	C4	38	DF	A8	19	E9	FC	90	54	C7	3A	22	92
D	C5	73	26	FB	44	15	A1	9F	2A	88	D2	BC	21	D4	FA	CB
E	A3	79	11	50	80	ED	1B	62	20	96	6B	AC	70	7B	C8	0A
F	EE	A9	4D	0D	E6	0F	60	AF	66	7E	37	5E	9E	74	CE	B1

5 Security and performance analysis

5.1 Key space

As a symmetric encryption algorithm, the key space is calculated from two aspects: (1) the hash value of 256-bit, and (2) the two initial conditions of 2D-IDHCM $(x_1,y_1,r)_s,(x_1,y_1,r)_{\oplus} \in (0,2^n\text{-}1]$, where $n \in \{8,16\}$. Therefore, the total key space can reach $S = 2^{128} \times 2^{16\times 6} = 2^{224} >> 2^{100}$, which is large enough to resist exhaustive attacks.

5.2 Key sensitivity analysis

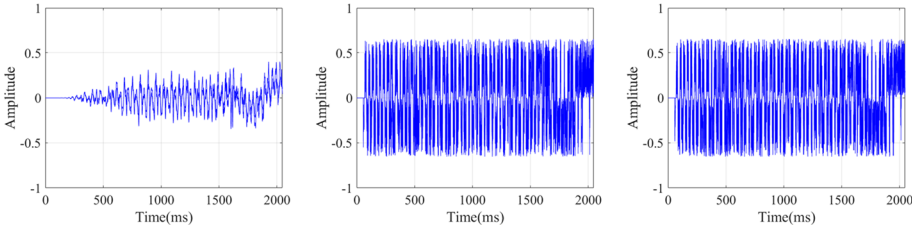
The sensitivity of chaotic maps to initial values can ensure the security of encryption algorithms in the face of brute force attacks. The initial condition of 2D-IDHCM is derived from true random environmental noise and is one-time one-key, even encrypting two identical plaint audios can produce two completely different ciphers.

Here, we randomly flip 1-bit of the hash value, and the corresponding decryption results of the Sample1.wav and the Sample2.wav are shown in Figs. 12 and 13 respectively, which demonstrated that the audio is undistinguished after decryption with the wrong keys, and the attacker cannot recognize the useful information. Hence, the algorithm is sensitive to tiny modification to keys.

As shown in Table 3, K represents the original key, K_1 and K_2 represent minor modifications to K respectively.

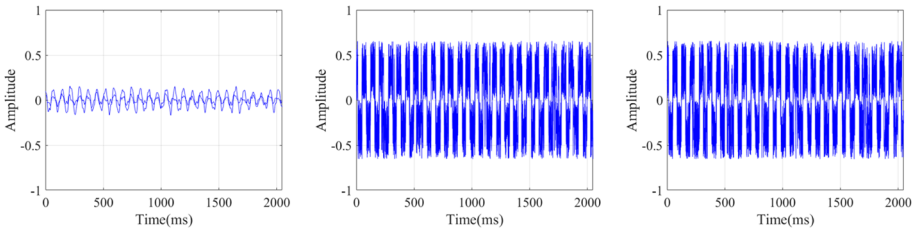
5.3 Histogram analysis

The histogram can illustrate the sample values of the audio signal and the frequency at which they appear in the audio signal, and the more uniform the distribution, the better the



(a) Decrypted audio with K (b) Decrypted audio with K_1 (c) Decrypted audio with K_2

Fig. 12 Key sensitivity testing (Sample1.wav)



(a) Decrypted audio with K (b) Decrypted audio with K_1 (c) Decrypted audio with K_2

Fig. 13 Key sensitivity testing (Sample2.wav)

Table 3 The keys in hexadecimal format

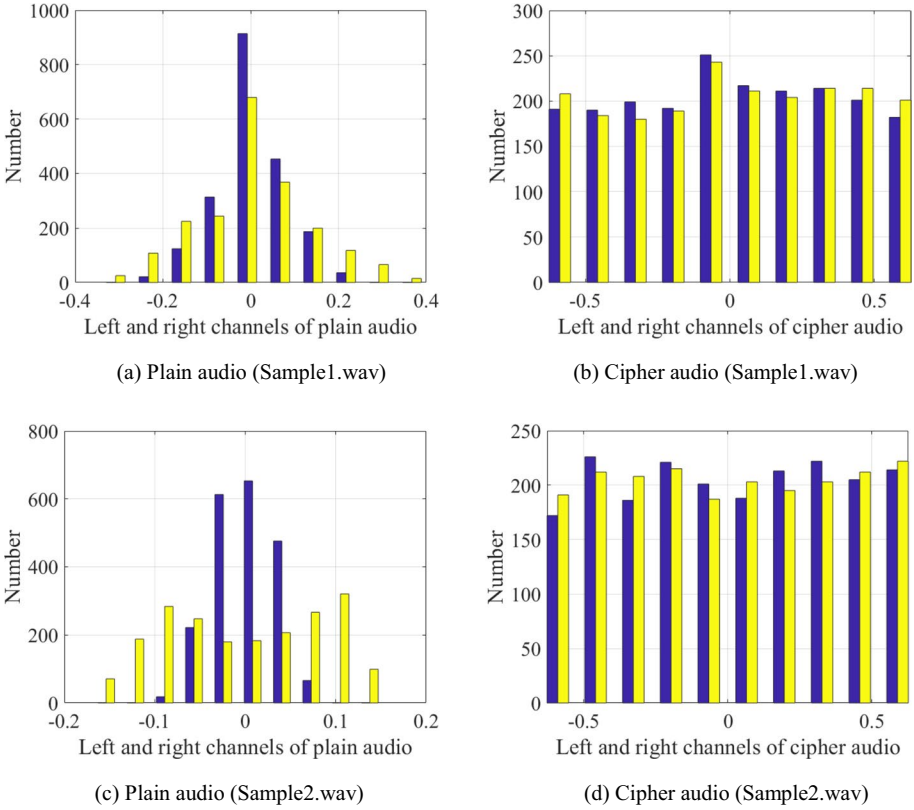
Type	Hash values
K	F3E614446B960FF91961ECDD1B0A2851009376AC89CD5B4E2D16ECBC9BE1E691
K_1	**E**3E614446B960FF91961ECDD1B0A2851009376AC89CD5B4E2D16ECBC9BE1E691
K_2	F3E614446B960FF91961ECDD1B0A2851009376AC89CD5B4E2D16ECBC9BE1E69**2**

encryption effect. To resist statistical attacks, the histogram distribution of cipher audio should be as uniform as possible [26]. Here the histograms of left and right channels of audio Sample1.wav and Sample2.wav are shown in Fig. 14, the results demonstrated that distribution of the left and right channels in the cipher audio is uniform, so the scheme can resist statistical attacks.

5.4 Correlation analysis

There is a high correlation between adjacent samples of the original audio. A good audio encryption algorithm should be able to reduce this correlation, the smaller the correlation coefficient between plain audio and its cipher, i.e., the similarity between them is worse, the more able to resist attacks [27].

We select four audio samples and test the correlation between the left channel (L) and the right channel (R) of each sample. The comparison of the proposed audio algorithm



(a) Plain audio (Sample1.wav)

(b) Cipher audio (Sample1.wav)

(c) Plain audio (Sample2.wav)

(d) Cipher audio (Sample2.wav)

Fig. 14 Histograms of the plain audio and cipher audio

with other algorithms is given in Table 4, where the subscripts P and C represent plain audio and cipher audio respectively, from which we can infer that the correlation in cipher is significantly decreased [28].

5.5 Differential attack

Differential attack refers to the attacker observing the difference between the original cipher audio and the new cipher audio by flipping the original audio 1-bit. From Table 5 we can infer that both the number of pixels change rate (NPCR) and the unified average changing

Table 4 The correlation between two channels and comparison with other algorithms

	Sample1.wav	Sample2.wav	Sample3.wav	Sample4.wav	Ref. [11]	Ref. [29]	Ref. [30]	Ref. [32]
L_P/R_P	0.501263	0.467471	0.263390	0.457578	0.692971	N/A	N/A	N/A
L_C/R_C	-0.003449	-0.003019	0.001091	-0.002490	0.005224	0.0049	0.0031	0.0020
L_P/L_C	-0.003481	-0.006837	0.000450	-0.005283	-0.001067	N/A	N/A	N/A
R_P/R_C	0.000923	0.005472	-0.001931	0.003435	-0.001229	N/A	N/A	N/A

Table 5 NPCR and UACI between ciphers and compared with other algorithms

Audios	NPCR(%)	UACI(%)	Audios	NPCR(%)	UACI(%)
Sample1.wav	99.6124	33.4176	Sample3.wav	99.5705	33.3435
Sample2.wav	99.5880	33.4227	Sample4.wav	99.5903	33.4256
Ref. [9]	99.9800	30.2400	Ref. [31]	99.9900	33.3100
Ref. [12]	99.6052	33.4860	Ref. [32]	99.9995	33.5599
Ref. [27]	99.6040	33.3300	Ref. [33]	99.9000	33.5200

Table 6 Comparison of the values of MSE and PSNR obtained by this algorithm with other algorithms

Audios	Sample1.wav	Sample2.wav	Sample3.wav	Sample4.wav	Ref. [11]	Ref. [27]	Ref. [29]	Ref. [31]
MSE	0.3082	0.3046	0.3870	0.3068	N/A	N/A	0.3263	N/A
PSNR	3.9278	4.7965	3.0925	4.2594	4.5299	5.1042	10.770	10.6707

intensity (UACI) are all close to their ideal values of 99.6094070% and 33.4635070% [31]. Therefore, the algorithm can resist the differential attack.

5.6 Mean square error (MSE) and Peak signal-to-noise ratio (PSNR)

MSE is the mean square error between original signal and encrypted noisy signal. PSNR is the ratio between the highest achievable power of a signal and the noise power that destroys its characterization [12]. PSNR is used to determine audio quality, a lower PSNR value indicates that the encryption algorithm is robust. Therefore, a good encryption algorithm should reduce the PSNR value of scrambled audio.

As shown in Table 6, we can infer that the encryption algorithm has the ability to resist attacks.

5.7 Information entropy

Information entropy is an index of the randomness of audio samples, which can be used to test the randomness of input information [34]. After encryption, information entropy should increase. The ideal value of information entropy is 8. The results in Table 7 demonstrate that the information entropy of the ciphertext is close to 8. Hence, the scheme is able to resist entropy analysis.

In summary, in Section 5, we conduct performance and security analysis in many aspects, and compare with other algorithms to verify that the proposed method has high security. However, due to the finite precision effect of computer, the audio encryption algorithm based on chaotic map in digital domain can cause dynamic degradation [12, 26]. Therefore, in order to resist dynamic degradation, we designed an audio encryption algorithm based on 2D-IDHCM over GF(2^n).

Table 7 Comparison of information entropy analysis with other algorithms

Audios	Ciphered audio	Audios	Ciphered audio
Sample1.wav	7.9908	Sample3.wav	7.9906
Sample2.wav	7.9912	Sample4.wav	7.9907
Ref. [10]	7.9990	Ref. [27]	7.9987
Ref. [26]	7.9980	Ref. [29]	7.9966

In this paper, we first apply 2D-IDHCM to audio signal encryption. Compared with chaotic map in digital domain, the advantage is that there is no need to convert floating-point numbers to integers when scrambling audio sequences, which improves the efficiency of the algorithm.

6 Conclusion

To ensure the security transmission of audio signals in instant messaging tools, we proposed a chaos-based dual-channel one-time one-key audio encryption scheme. First, we constructed a 2D-IDHCM over GF(2^n) with ergodicity and large Lyapunov exponents. More importantly, the 2D-IDHCM solves the dynamic degradation of chaotic map in digital domain caused by truncation error and rounding error, and the generated sequences have high nonlinearity. Secondly, based on 2D-IDHCM, we designed two keyed strong S-Boxes without fixed point, reverse fixed points and short period rings. Further, the algorithm consists of two processes: confusion and diffusion. (1) We applied the integer sequences generated by 2D-IDHCM to XOR the elements in the two channels. (2) The positions of elements in the left and right channels were confused by two keyed strong S-Boxes, and there is no need to convert floating point numbers to integers when scrambling the audio sequences, which improves the efficiency. In addition, the initial condition of 2D-IDHCM relies on the hash value of true random environmental noise, which can help generate different initial conditions and improve the security of encryption algorithm. Experimental results demonstrated that the proposed audio encryption scheme is sensitive to the initial condition, and the total key space can reach 2^{224}, which can resist exhaustive attacks. In addition, NPCR, UACI and the information entropy are close to their ideal values. Therefore, it is suitable for real-time secure communication. The shortcoming of the proposed method is that the nonlinearity of S-Box is not very high. In the future, we can improve the nonlinearity of S-Box by combining chaos with algebraic structure to enhance the security. Then, we will further study the practical application of chaos-based audio encryption in instant messaging.

Acknowledgements This research is supported by the Natural Science Foundation of Shandong Province (No. ZR2022MF232), and the Teaching and Research Program of University of Jinan (No. J2203).

Data availability Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Suryadi MT, Satria Y, Boyke M (2023) Digital audio protection with confusion and diffusion scheme using double-scroll chaotic function. J Hunan Univ Nat Sci 50(5):45–52
2. Joshi AB, Gaffar A (2023) A technique for securing digital audio files based on rotation and XOR operations. Soft Comput 28:5523–5540
3. Jana S, Dutta N, Maji A K et al (2022) A novel time-stamp-based audio encryption scheme using sudoku puzzle[C]//Proceedings of International Conference on Frontiers in Computing and Systems: COMSYS 2021. Springer Nature Singapore, Singapore, pp 159–169
4. Kumar A, Dua M (2023) Audio encryption using two chaotic map based dynamic diffusion and double DNA encoding. Appl Acoust 203:109196
5. Shah D, Shah T, Ahamad I et al (2021) A three-dimensional chaotic map and their applications to digital audio security. Multimed Tools Appl 80:22251–22273
6. Fu SM, Cheng XF, Liu J (2023) Dynamics, circuit design, feedback control of a new hyper chaotic system and its application in audio encryption. Sci Rep 13(1):19385
7. Wang X, Liu P (2021) A new full chaos coupled mapping lattice and its application in privacy image encryption. IEEE Trans Circuits Syst I Regul Pap 69(3):1291–1301
8. Wang M, Liu H, Zhao M (2022) Bit-level image encryption algorithm based on random-time S-Box substitution. Eur Phys J Special Topics 231(16):3225–3237
9. Liu H, Kadir A, Li Y (2016) Audio encryption scheme by confusion and diffusion based on multi-scroll chaotic system and one-time keys. Optik 127(19):7431–7438
10. Wu R, Gao S, Wang X et al (2022) AEA-NCS: An audio encryption algorithm based on a nested chaotic system. Chaos, Solitons Fractals 165:112770
11. Rahul B, Kuppusamy K, Senthilrajan A (2023) Chaos-based audio encryption algorithm using biometric image and SHA-256 hash algorithm. Multimedia Tools Appl 82(28):43729–43758
12. Liu H (2023) Audio block encryption using 3D chaotic system with adaptive parameter perturbation. Multimed Tools Appl 82:27973–27987
13. Ben Farah MA, Guesmi R, Kachouri A et al (2020) A new design of cryptosystem based on S-Box and chaotic permutation. Multimed Tools Appl 79(27–28):19129–19150
14. Si Y, Liu H, Zhao M (2023) Constructing keyed strong S-Box with higher nonlinearity based on 2D hyper chaotic map and algebraic operation. Integration 88:269–277
15. Liu H, Wang X, Zhao M et al (2022) Constructing strong S-Box by 2D chaotic map with application to irreversible parallel key expansion. Int J Bifurcation Chaos 32(11):2250163
16. Fan C, Ding Q (2023) A novel algorithm to analyze the dynamics of digital chaotic maps in finite-precision domain. Chin Phys B 32(1):010501
17. Luo Y, Liu Y, Liu J et al (2021) Counteracting dynamical degradation of a class of digital chaotic systems via unscented Kalman filter and perturbation. Inf Sci 556:49–66
18. Fan C, Ding Q (2021) Analysis and resistance of dynamic degradation of digital chaos via functional graphs. Nonlinear Dyn 103(1):1081–1097
19. Wang Q, Yu S, Guyeux C, Bahi JM, Fang X (2014) Theoretical design and circuit implementation of integer domain chaotic systems. Int J Bifurcation Chaos 24(10):1450128
20. Bashier E, Jabeur TB (2021) An efficient secure image encryption algorithm based on total shuffling, integer chaotic maps and median filter. J Internet Serv Inform Secur 11(2):46–77
21. Si Y, Liu H, Chen Y (2021) Constructing keyed strong S-Box using an enhanced quadratic map. Int J Bifurcation Chaos 31(10):2150146
22. Cao Y, Liu H, Xu D (2023) Constructing a nondegenerate 2D integer-domain hyperchaotic map over GF(2^n) with application in parallel hashing. Int J Bifurcation Chaos 33(15):2350181
23. Zhao M, Liu H (2022) Construction of a nondegenerate 2D chaotic map with application to irreversible parallel key expansion algorithm. Int J Bifurcation Chaos 32(6):2250081
24. Cao W, Mao Y, Zhou Y (2020) Designing a 2D infinite collapse map for image encryption. Signal Process 171:107457
25. Si Y, Liu H, Chen Y (2022) Constructing a 3D exponential hyper chaotic map with application to PRNG. Int J Bifurcation Chaos 32(7):2250095
26. Demirtaş M (2023) A lossless audio encryption method based on Chebyshev map. Orclever Proc Res Dev 2(1):28–38
27. Albahrani EA, Alshekly TK, Lafta SH (2023) New secure and efficient substitution and permutation method for audio encryption algorithm. J Supercomput 79(15):16616–16646
28. Adhikari S, Karforma S (2021) A novel audio encryption method using Henon-Tent chaotic pseudo random number sequence. Int J Inf Technol 13(4):1463–1471

29. Shah D, Shah T, Jamal SS (2020) Digital audio signals encryption by Mobius transformation and Hénon map. Multimedia Syst 26:235–245
30. Aziz H, Gilani SMM, Hussain I et al (2021) A noise-tolerant audio encryption framework designed by the application of S8 symmetric group and chaotic systems. Math Probl Eng 2021:5554707
31. Liu H, Liu J, Ma C (2023) Constructing dynamic strong S-Box using 3D chaotic map and application to image encryption. Multimed Tools Appl 82(16):23899–23914
32. Naskar PK, Bhattacharyya S, Chaudhuri A (2021) An audio encryption based on distinct key blocks along with PWLCM and ECA. Nonlinear Dyn 103:2019–2042
33. Alanazi AS, Munir N, Khan M et al (2023) A novel design of audio signals encryption with substitution permutation network based on the Genesio-Tesi chaotic system. Multimed Tools Appl 82(17):26577–26593
34. Wang M, Liu H, Zhao M (2023) Construction of a non-degeneracy 3D chaotic map and application to image encryption with keyed S-Box. Multimed Tools Appl 82:34541–34563

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.